

Introduction to Statistics and Probability

Assignment # 2

2025-2026

Exercise 1 – Let X_1 and X_2 be independent exponentially distributed r.v. with parameters λ_1 and λ_2 respectively.

Let :

$$N = \begin{cases} 1 & \text{if } X_1 < X_2, \\ 2 & \text{if } X_1 \geq X_2. \end{cases}$$

$U = \min\{X_1, X_2\}$, $V = \max\{X_1, X_2\}$, and $W = V - U = |X_1 - X_2|$.

Show that

1. $P(N = 1) = \lambda_1/(\lambda_1 + \lambda_2)$ and $P(N = 2) = \lambda_2/(\lambda_1 + \lambda_2)$.
2. $P(U > t) = \exp\{-(\lambda_1 + \lambda_2)t\}$ for $t \geq 0$.
3. N and U are independent random variables.
4. $P(W > t | N = 1) = \exp\{-\lambda_2 t\}$ and $P(W > t | N = 2) = \exp\{-\lambda_1 t\}$ for $t \geq 0$.
5. U and $W = V - U$ are independent random variables

Exercise 2 – To describe a bivariate r.v. $X = (X_1, X_2)$ random variable, the vectorial notation is used. We say that a random vector $X = [X_1 \ X_2]'$ has the bivariate normal distribution $\mathcal{N}(\mu, \Sigma)$ where $\mu = [\mu_1 \ \mu_2]'$ and $\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix}$, a symmetric positive definite matrix and we denote

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix}\right).$$

iff its probability density function is expressed as :

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sqrt{\det(\Sigma)}} \exp\left\{-\frac{1}{2}(x - \mu)' \Sigma^{-1}(x - \mu)\right\},$$

where $x_1 \in \mathbb{R}$, $x_2 \in \mathbb{R}$, and $\det(\Sigma)$ is the determinant of the matrix Σ .

1. Show that the p.d.f. can be expressed as :

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]\right\},$$

where $\rho = \sigma_{12}/\sqrt{\sigma_{11}\sigma_{22}}$.

2. Find the marginal p.d.f.'s of X_1 and X_2 .
3. Find the conditional p.d.f. of X_1 given $X_2 = x_2$, and the conditional p.d.f. of X_2 given $X_1 = x_1$.

4. Show that if $\rho = 0$ then X_1 and X_2 are independent r.v.'s.
5. Let (X_1, X_2) be a bivariate r.v. with joint p.d.f.

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi} e^{-(x_1^2 + x_2^2)/2} (1 + x_1 x_2 e^{(x_1^2 + x_2^2 - 2)/2}), \text{ for } x_1 \in \mathbb{R} \text{ and } x_2 \in \mathbb{R}.$$

6. Show that $f_{X_1, X_2}(x_1, x_2)$ is a p.d.f.
7. Find and identify the marginal p.d.f.'s of X_1 and X_2 .
8. Is $f_{X_1, X_2}(x_1, x_2)$ a bivariate normal distribution?
Write a conclusion.

Exercise 3 – Let $(X_1, X_2, \dots, X_n)$ be a n r.v.'s *i.i.d.*, each with probability density function :

$$f(x | \theta) = \theta(\theta + 1) x^{\theta-1} (1 - x), \quad 0 < x < 1, \theta > 0.$$

I. Estimation

1. Check that $f(x | \theta)$ is a probability density function.
2. Use the method of moments to find an estimator of θ .
3. Find the maximum likelihood estimator of θ .

II. Simulation with Python

1. A random variable Y has a beta distribution with parameters (α, β) iff its p.d.f has the following expression :

$$f_Y(y) = \frac{1}{B(\alpha, \beta)} y^{\alpha-1} (1 - y)^{\beta-1}, \text{ where } B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}, \alpha, \beta > 0.$$

Show that $f(x | \theta)$ is the p.d.f. of a beta distribution ; identify the parameters.

Hint : $\forall \eta > 0, \Gamma(\eta + 1) = \eta \Gamma(\eta)$.

2. Simulate M n -sample from $f(x | 0.37)$.
For each sample compute $MME(\theta)^{(i)}$ and $MLE(\theta)^{(i)}$, resp. the method of moments estimator and the maximum likelihood estimator of θ for the i^{th} sample, $i = 1, \dots, M$.
3. Compute the means of the estimates. Comment the results.
4. Construct the histograms of $MME(\theta)^{(i)}$ and $MLE(\theta)^{(i)}$, $i = 1, \dots, M$.
Compare and comment.